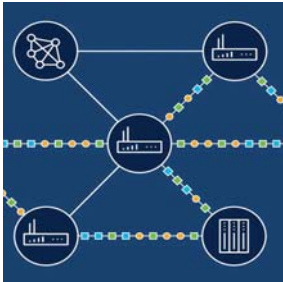


A 2.3.4 Static vs. Dynamic Routing (HL Only)



A2.3.4 Explain how static routing and dynamic routing move data across local area networks. (HL only)

Explore static and dynamic routing in networks. Learn how routers direct traffic manually (static) or automatically (dynamic). Discover the pros and cons and key factors like network size and complexity influencing routing choices.

Introduction

Data must travel from a source device to a destination device in computer networks, often traversing multiple networks. This is where routing comes in. Routing is the process of selecting paths through a network to send data.

This section explores two fundamental routing approaches: **static routing** and **dynamic routing**. We'll explore their processes, advantages, disadvantages, and factors influencing their suitability for different network scenarios.

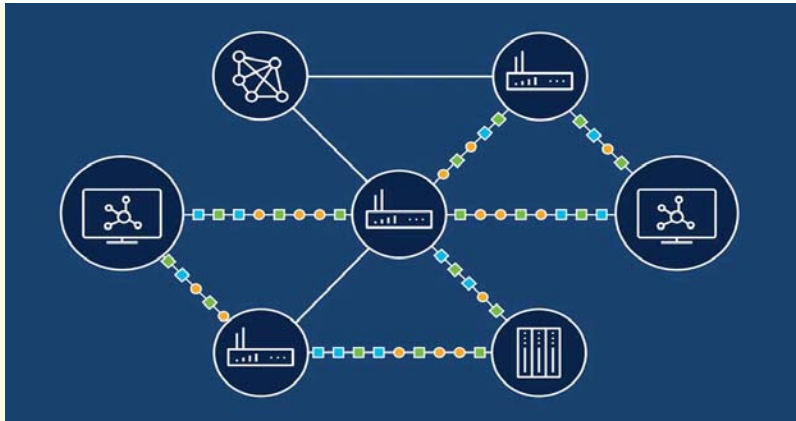
Static Routing: The Manual Approach

Static routing involves manually configuring routing tables on routers. Network administrators define specific paths for data to take, essentially creating a fixed roadmap for network traffic.

How it works:

Manual Configuration: Administrators manually enter routing information into the router's routing table, specifying the destination network, the next hop router, and the associated interface.

Fixed Paths: Data packets follow these predefined paths regardless of network changes or congestion.



Advantages:

- **Predictability:** Network traffic follows predefined paths, ensuring consistent performance.
- **Security:** Administrators have granular control over network traffic flow, potentially enhancing security.
- **Efficiency:** Static routes can be optimised for specific network layouts, potentially reducing overhead.
- **Simplicity:** In small, simple networks, static routing can be easier to implement.

Disadvantages:

- **Lack of Adaptability:** Static routes don't adapt to network changes (e.g., link failures or congestion), which can lead to disruptions.
- **Maintenance:** Manual configuration and updates can be time-consuming, especially in larger networks.
- **Scalability:** Not ideal for large or complex networks due to the increased management overhead.

Dynamic Routing: The Adaptive Approach

Dynamic routing allows routers to learn and adapt to network changes automatically. Routers exchange routing information with each other using routing protocols, enabling them to build and maintain dynamic routing tables.

How it works:

Routing Protocols: Routers use protocols (e.g., RIP, OSPF, BGP) to communicate with each other and share information about connected networks and link costs.

Dynamic Routing Tables: Routers build and update their routing tables based on the information

received from other routers.

Adaptive Routing: Data packets are forwarded along the best available path, which can change in response to network conditions.

Advantages:

- **Adaptability:** Dynamic routing automatically adjusts to network changes, providing fault tolerance and improved performance.
- **Reduced Administration:** Requires less manual intervention, as routers automatically learn and update routes.
- **Scalability:** Suitable for large and complex networks, as the routing process is distributed among routers.

Disadvantages:

- **Complexity:** Dynamic routing protocols can be complex to configure and troubleshoot.
- **Resource Usage:** Routers consume resources (CPU, memory, bandwidth) to exchange routing information and calculate optimal paths.
- **Convergence Time:** It takes time for routers to converge on a consistent view of the network after a change, which can cause temporary routing loops or instability.

Factors to Consider: Configuration, Maintenance, Complexity, Resource Usage, Convergence, Scalability, Network Size

When choosing between static and dynamic routing, several factors must be considered:

Factor	Choice
Configuration	Static routing requires manual configuration, while dynamic routing relies on routing protocols.
Maintenance	Static routes require manual updates, while dynamic routing adapts automatically.
Complexity	Dynamic routing is generally more complex than static routing.
Resource Usage	Dynamic routing consumes more router resources.
Convergence	Dynamic routing takes time to converge after network changes.

Factor	Choice
Scalability	Dynamic routing is more scalable for larger networks.
Network Size	Static routing may be suitable for small, simple networks, while dynamic routing is better for larger, more complex networks.

Key Questions:

Q: What are the key differences between static and dynamic routing?

A: Static routing uses manual configuration and fixed paths, while dynamic routing uses routing protocols and adapts to network changes.

Q: In what situations would static routing be preferred over dynamic routing?

A: In small, simple networks where predictability and security are paramount, and administrative overhead is minimal.

Q: How do dynamic routing protocols enable routers to learn about network changes?

A: Routers exchange routing information with each other, sharing updates about connected networks and link costs.

Q: What is convergence time, and why is it an important consideration in dynamic routing?

A: Convergence time is the time it takes for routers to agree on the best paths after a network change. Slow convergence can lead to temporary routing loops or instability.

Q: How does the size and complexity of a network influence the choice between static and dynamic routing?

A: Static routing becomes increasingly difficult to manage in larger networks, while dynamic routing scales better due to its distributed nature.

Q: What are the potential security implications of using static routing?

A: If not properly configured, static routes can create vulnerabilities that attackers could exploit to redirect or intercept network traffic.

Q: How does dynamic routing contribute to the fault tolerance of a network?

A: If a link or router fails, dynamic routing protocols automatically recalculate routes to bypass the failed component, ensuring continued connectivity.

Quiz

Test your knowledge with this quick quiz.

1

Maintenance

network changes

complexity

Scalability

manually

Routing protocols

Convergence time

Routing tables

Resource usage

automatically

dynamic

Static routing

[Static routing] involves manually configuring routing tables on routers.

In **[dynamic]** routing, routers automatically learn and adapt to network changes.

[Routing tables] store information about network paths and destinations.

Static routes are configured **[manually]** by network administrators.

Dynamic routes are updated **[automatically]** by routing protocols.

[Routing protocols] enable routers to exchange routing information.

Dynamic routing offers better adaptability to **[network changes]**.

[Scalability] refers to the ability of a network to grow and handle increasing traffic.

[Maintenance] can be a significant overhead for static routing in large networks.

Dynamic routing protocols can increase **[complexity]** on routers.

[Convergence time] is the time it takes for routers to agree on the best paths after a network change.

[Resource usage] is a factor to consider when choosing between static and dynamic routing, as dynamic routing protocols consume more router resources.

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